



FINE JEWELLERY · EST. MMXXVI

OPALÉRA

Named for the opal — the one stone whose fire holds the colour of every gem: the blue of sapphire, the blush of rose gold, the light of diamond and the lustre of pearl. Eighty handcrafted pieces, one house — and a studio to make yours truly yours.



IFM2B10 · GROUP PROJECT · FINAL PRESENTATION · SEPTEMBER 2026

OPALÉRA — The House of Every Gem

A full-stack B2C fine-jewellery store · Quantum Coders · 80 pieces · 92 pages · Node.js backend · manager console · live at opalera.onrender.com

The Team



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Database creator



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Dataset: the 80-piece Kaggle jewellery database · necklaces, earrings & pendants · R 959 – R 6 799 · Source: github.com/Thriyi17/Opalera

Why OPALÉRA Exists

Buying jewellery online is a gamble — fake stones, no recourse, real pieces priced beyond a normal salary. OPALÉRA answers with a maison you can trust.



Certified authentic

Every piece carries its own certificate of authenticity; the maison refunds in full if a stone is ever misrepresented.



Honest prices

Direct-from-atelier pricing from R 959, VAT included, with the comparable boutique estimate and instalment framing on every piece.



The rating system

Star ratings, written patron reviews with a verified-purchase badge, and thumbs votes — visible before you pay.

What We Built

80

PIECES IN THE
DATABASE

92

HTML PAGES

24

API ENDPOINTS

2

USER ROLES

64

AUTOMATED CHECKS

0

NPM DEPENDENCIES



Node.js server

server.js — routes, script password hashing, httpOnly cookie sessions, server-side totals and stock rules, static hosting of the site



JSON database

data/db.json — users, sessions, products, orders, reviews, wishlists, carts, resets; seeded on first run; DATA_DIR for a persistent disk



Shared JS layer

opalera-data.js — catalogue helpers, the API client, the live catalogue mirror, order tracking; one script for every page



Deployed

GitHub → Render: every push to main redeploys
opalera.onrender.com — the same server, site and API

Every Feature Stands Complete



Website design

one gold-on-midnight system, 92 pages



Product catalog

service-fed, search · price bands · sort · tabs



Database

8 collections, 80 seeded products



Web service

24 JSON endpoints, cookie sessions



Product management

add · edit · retire · delete, live on the storefront



Shopping cart

quantities, stock caps, saved per account



Transactions

promo codes, 15% VAT, stock rule, 3 pay methods



Invoices

tax invoice per order, QR code, printable



User management

register · login · reset · patron & manager roles

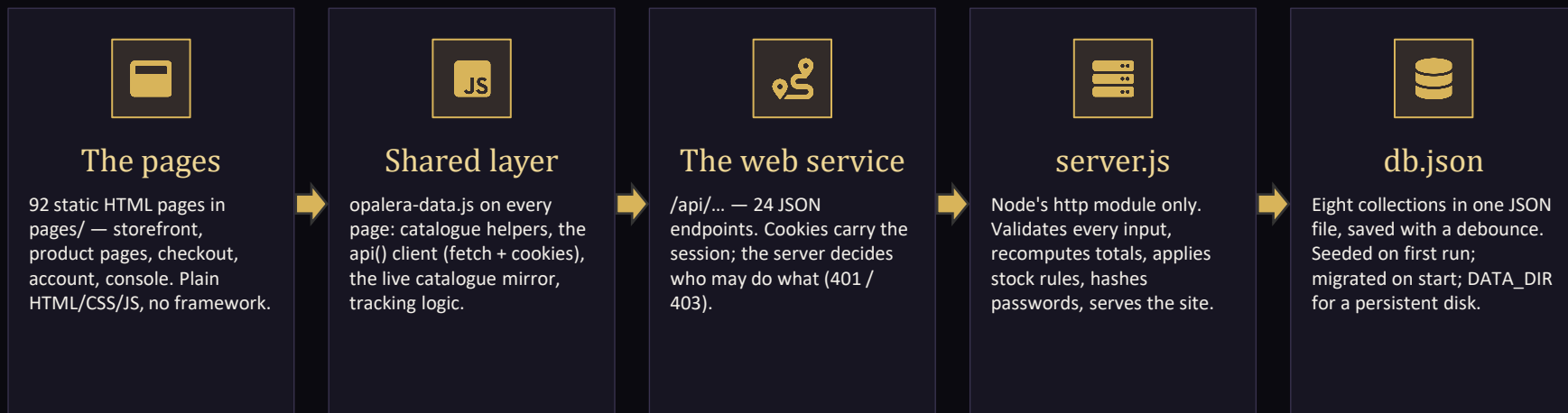


Reports

filters, distinct-products control, stock, users/day, 6 KPIs

BONUS · compare pieces side by side · live order-tracking timeline · QR code on every invoice · the Studio customiser

One Request, End to End



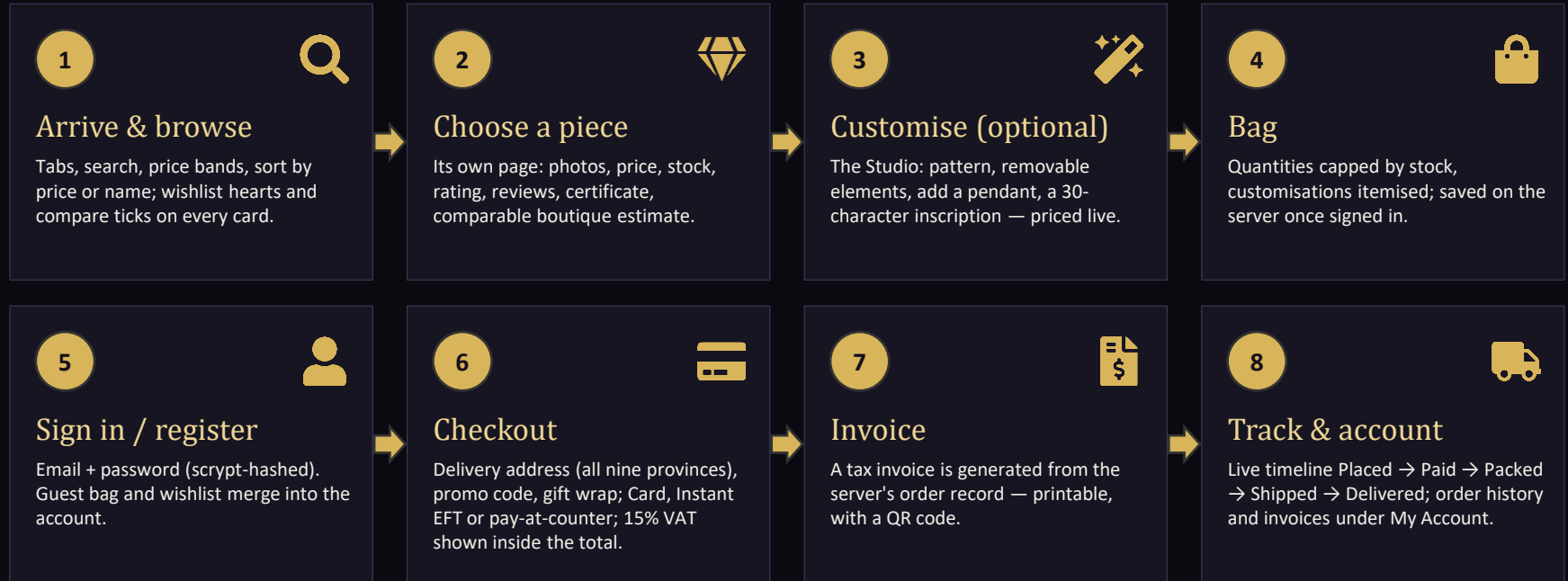
A patron opens a product page the page paints from the bundled catalogue instantly, then GET `/api/products` patches live price, stock and retired flags in place

A patron checks out POST `/api/orders` — the server recomputes item totals, discount and VAT, refuses over-ordering, deducts stock, clears the cart, stores the order

A manager edits a piece PUT `/api/products/:id` (manager cookie) — saved to `db.json`; every storefront tab and product page shows the change on its next fetch

Deployment git push to main → Render rebuilds and restarts the same server; the database re-seeds from the repo's copy

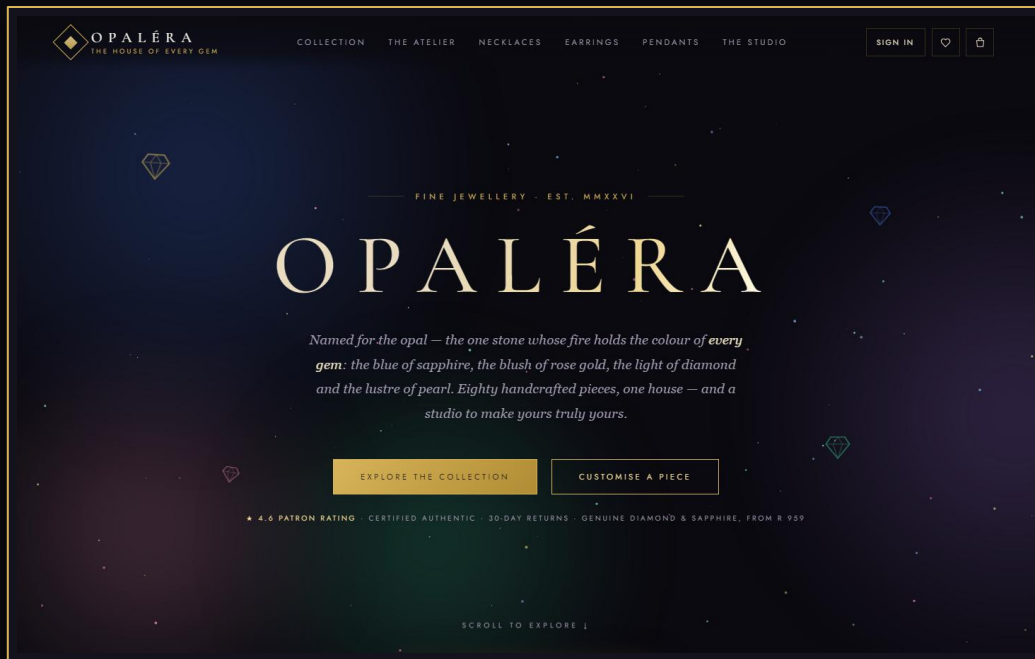
From Arrival to Delivered, in Eight Steps



Arrival at the Maison

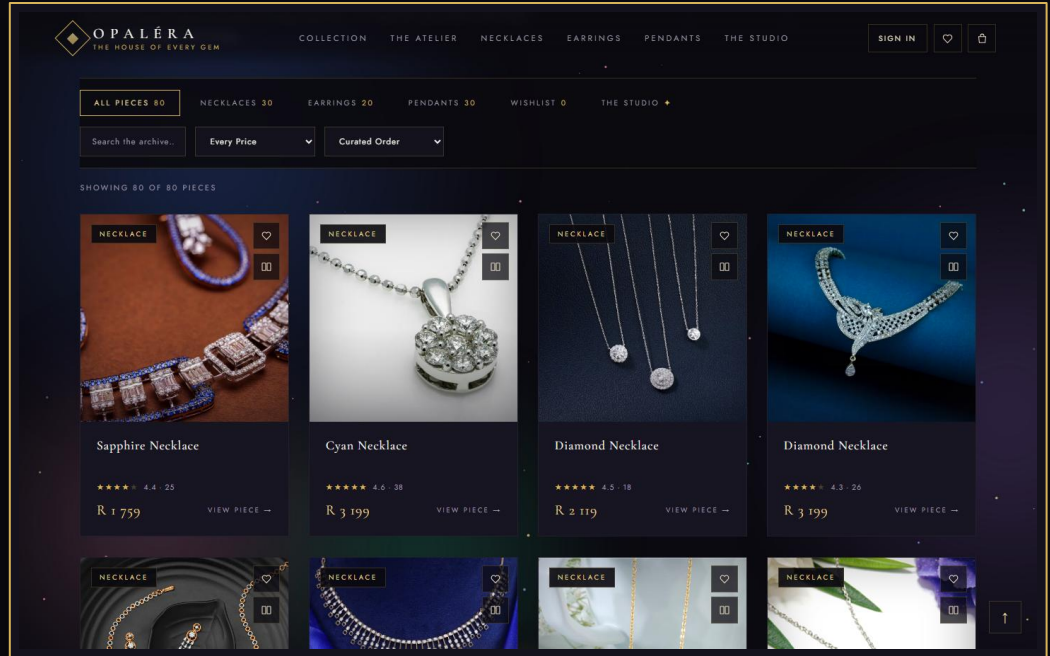
- ◆ Animated aurora, sparkle field and floating gems behind every page
- ◆ One gold-on-midnight design system across all 92 pages
- ◆ Trust strip on arrival: rating · authenticity · returns · from R 959
- ◆ Sign-in, wishlist and bag in the nav — the counts go live the moment a patron signs in

DESIGN · DONE



The Collection — Fed by the Service

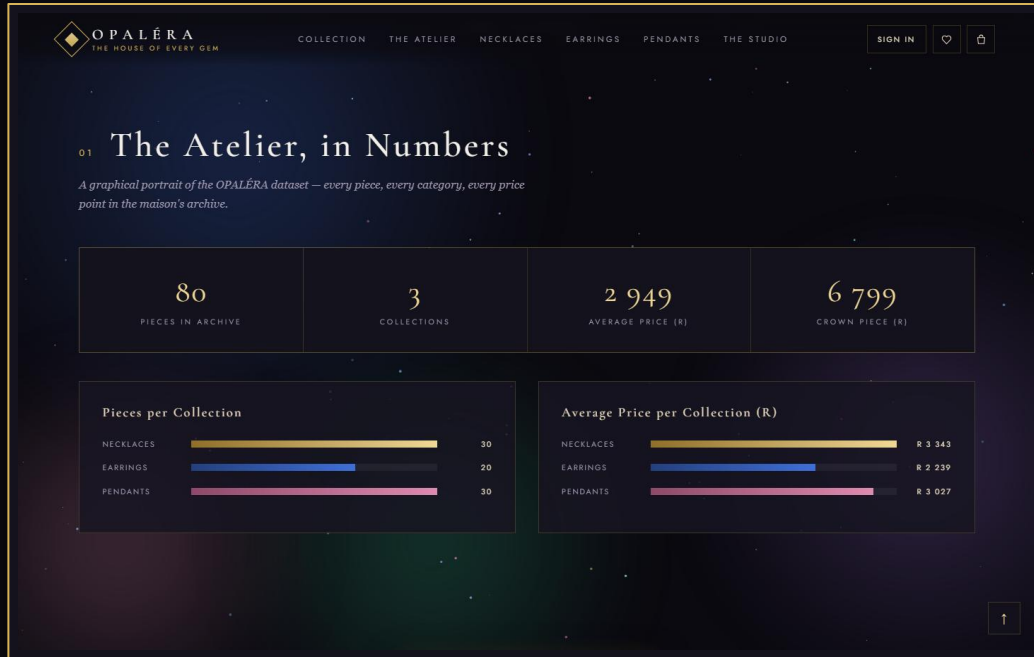
- ◆ The catalogue is retrieved from the database through GET /api/products — prices, photos, stock and retired pieces update live
- ◆ Category tabs with live counts, search, four price bands and sorting by price or name (the two sort criteria)
- ◆ “Showing 80 of 80 pieces” honesty line; wishlist hearts and a compare tick on every card
- ◆ Each card opens the piece's own page



CATALOG · DONE

SERVICE · DONE

The Atelier, in Numbers



- ◆ Live stat counters computed from the catalogue: 80 pieces, 3 collections, average price R 2 949, crown piece R 6 799
- ◆ Pieces per collection and average price per collection, coloured by gem accent
- ◆ Counts animate in as the visitor scrolls
- ◆ Everything here recomputes if the manager adds or retires pieces

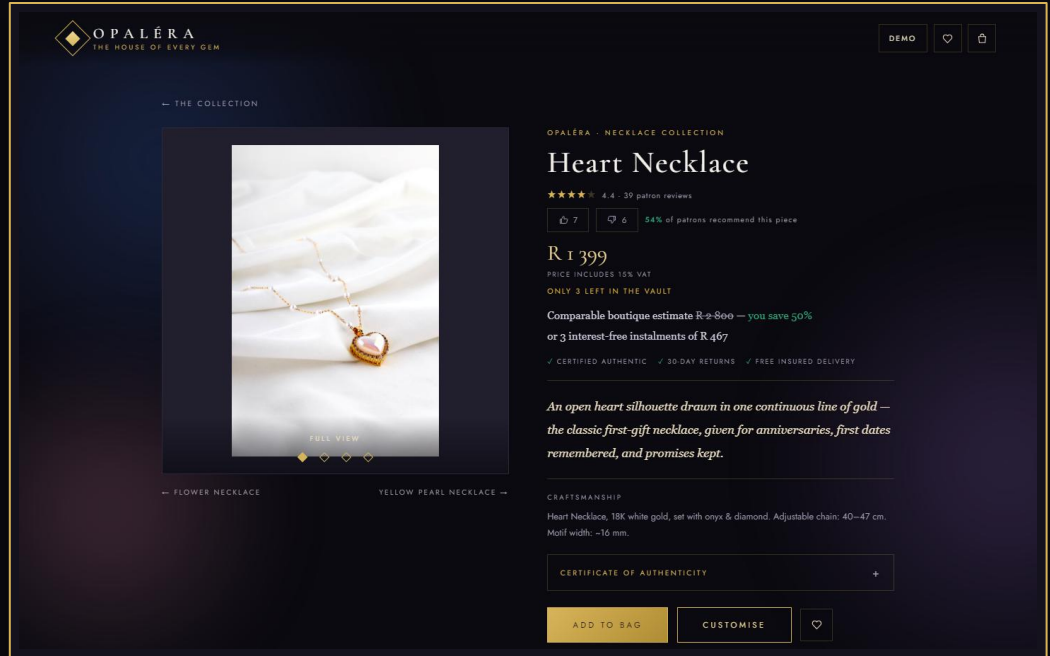
DESIGN · DONE

The Product Page

- ◆ Hand-written description for all 80 pieces, plus craftsmanship specs
- ◆ Live stock from the database — “Only 3 left in the vault”
- ◆ Stars, thumbs votes and “% recommend”; reviews with the verified-purchase badge
- ◆ Comparable boutique estimate, instalments, certificate of authenticity
- ◆ Manager-added pieces get a page automatically (product.html?id=...)

INFORMATION · DONE

RATINGS · LIVE



Ratings You Can Trust



Star ratings

Every piece shows an average out of 5 and the review count. Patron reviews posted through the site are blended into the seeded baseline, so the number moves as real reviews arrive.



Written reviews

Signed-in patrons write a review (1–5 stars, up to 240 characters) on any piece — `POST /api/reviews`. A patron may delete their own; a manager may delete any.



Verified purchase

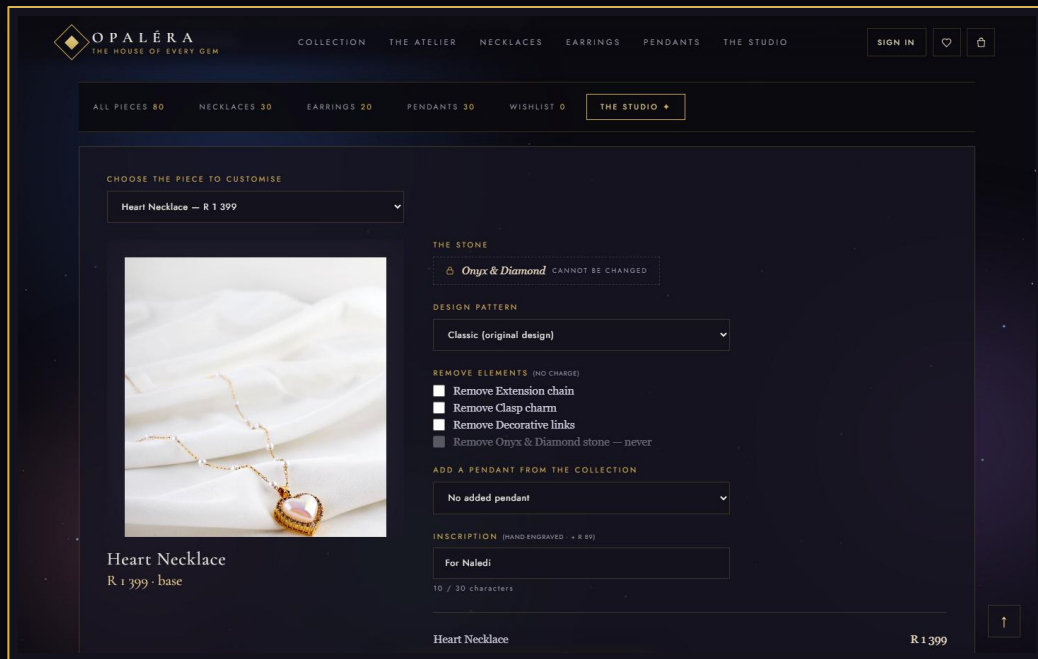
When a review is posted the server looks through that patron's orders; if the piece is there the review is stamped VERIFIED. Nobody can fake the badge from the browser.



Thumbs & “% recommend”

One vote per browser per piece, tap again to undo. The console's Engagement tab ranks pieces by recommend share and shows the split by collection.

The Studio — Customisation, With Rules



- ◆ Choose any piece, then: design pattern, removable elements, add a pendant from the collection, a hand-engraved inscription (+ R 89)
- ◆ The stone is never changed — locked on screen; signature pieces allow the inscription only
- ◆ 30-character inscription with a content filter
- ◆ Priced live; the customised piece is carried to the bag, the invoice and the order record

BONUS

Checkout: Quantities, Rules & VAT

- ◆ Bag with quantities, capped by real stock; customisations itemised; saved per account on the server
- ◆ Promo code (GEM10 / WELCOME10), complimentary gift wrap, delivery address with all nine provinces
- ◆ Card (simulated, never stored), Instant EFT or pay-at-counter — totals recomputed on the server so the client can't tamper
- ◆ Stock is deducted on the server the moment the order is placed

CART · DONE

TRANSACTIONS · DONE

The screenshot displays the 'SECURE CHECKOUT' page for 'OPALÉRA THE HOUSE OF EVERY GEM'. The 'Payment' step is active, with progress indicators for 'YOUR BAG', 'DELIVERY & PAYMENT', and 'CONFIRMATION & INVOICE'. The page is divided into three main sections: 'Your Order', 'Delivery Address', and 'Payment'.

Your Order

Item	Quantity	Price
Heart Necklace	1	R 1 399
Sapphire Necklace	2	R 3 518

Promo Code (TRY GEM10) [APPLY]

☐ Wrap this order as a gift (complimentary)

Items: R 4 917
VAT included (15%): R 641
Delivery (insured, tracked): Free

Total due: R 4 917

Delivery Address

RECIPIENT FULL NAME: Demo Patron

STREET ADDRESS: 12 Rosebank Avenue

CITY / TOWN: [] POSTAL CODE: []

PROVINCE: Gauteng PHONE: 071 234 5678

Payment — for Demo Patron

Eight Rules, Enforced Where It Counts



Promo codes

GEM10 / WELCOME10 take 10% off the items. The discount is capped at the items total and re-applied server-side.



15% VAT inside the price

Prices are VAT-inclusive; VAT = total $\times$ 15 / 115 is shown on the bag, the checkout and the invoice.



No over-ordering

Server rule: an order for more units than the vault holds is refused — “Only 18 of Sapphire Necklace left”. Retired pieces are refused too.



Quantity capped by stock

The bag's + button stops at the live stock figure; stock is deducted on the server at checkout.



Signature pieces

The stone is never customisable; signature pieces allow an inscription only.



Inscription filter

30 characters, hand-engraved at + R 89, with a content filter on the text.



Delivery & collection

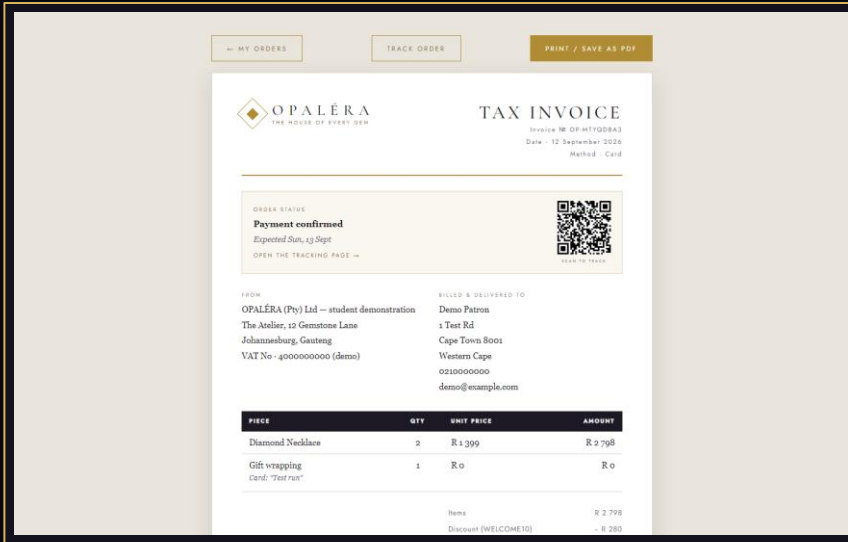
Delivery address requires one of the nine provinces; pay-at-counter orders skip the address and are reserved, not paid.



Verified reviews

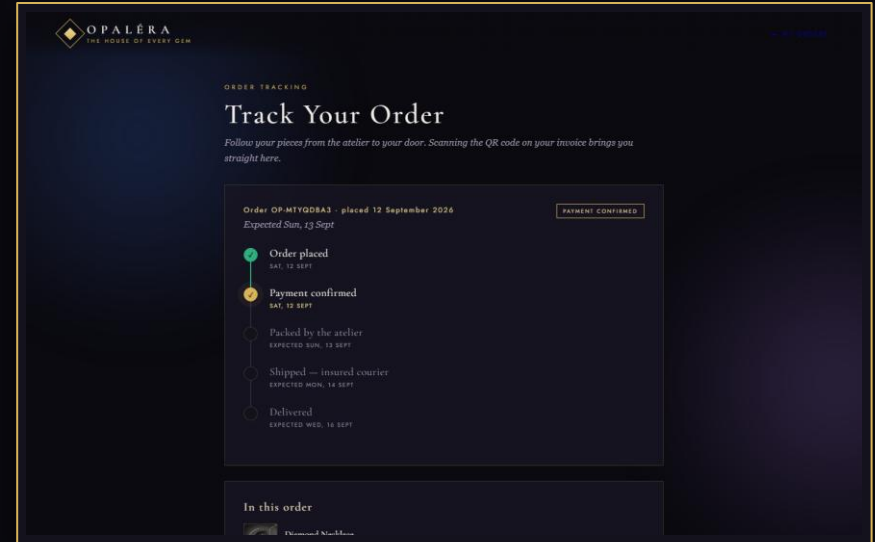
A review is marked verified only if the server finds the piece in that patron's own orders.

A Tax Invoice — and Where Your Order Is



Tax invoice · generated from the server's order record: letterhead, invoice No, date, method, line items with inscriptions, discount and VAT broken out, print / save as PDF.

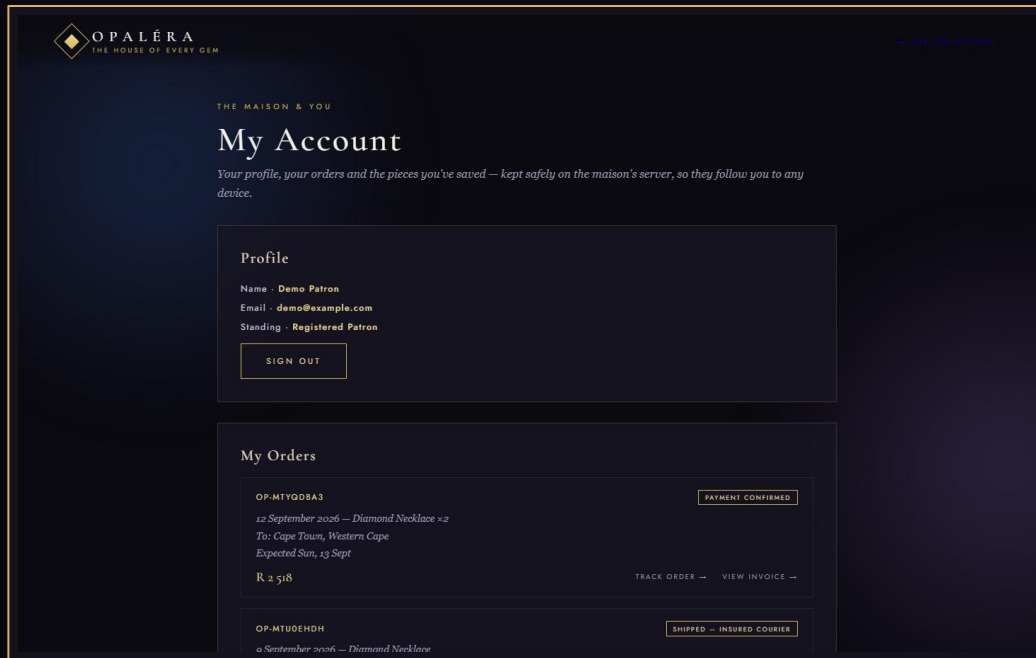
INVOICES · DONE



Tracking timeline · Placed → Paid → Packed → Shipped → Delivered (a collection path for pay-at-counter), advancing with time; the QR on the invoice opens it.

BONUS

Three Kinds of Visitor, Three Doors



- ◆ Guest — browses, searches, votes, reads reviews
- ◆ Patron — registers (bcrypt-hashed password, httpOnly cookie session); wishlist and bag saved on the server; orders, verified reviews, order history & invoices
- ◆ Manager — a separate console session (role “manager” on the server); patron accounts are refused; managers can also remove a patron account
- ◆ Forgot-password: one-time 6-digit code, single use, 15-minute expiry; show-password eye on every field

REGISTRATION · DONE

LOGIN · DONE

ROLES · DONE

Sign In, Stay In, Reset Safely



Passwords are never stored

Each account keeps a random salt and an script hash; sign-in compares hashes in constant time. Minimum 6 characters; emails validated and de-duplicated.



Cookie sessions

A random 32-byte token in an httpOnly, SameSite cookie — invisible to page scripts — valid 30 days. Log out destroys it on the server.



Two doors, two cookies

Patron sessions and manager sessions use different cookies, so a customer signed in on the storefront and the manager in the console never collide.



Forgot password

Request a 6-digit code (email delivery is simulated for the demo), 15-minute expiry, 5 attempts, single use; every other session is signed out on reset.



Bags that follow you

A guest's bag and wishlist are merged into the account on sign-in and pushed to the server; the pages mirror the signed-in patron locally for instant paint.



Roles on the server

patron / manager stored per account; manager-only routes answer 401 without a session and 403 to a patron. The backend folder is never served over HTTP.

One Server, Eight Collections, 24 Endpoints



users

email · names · script salt+hash · role
· createdAt



sessions

token → email · expiry (patron &
manager scopes)



products

80 pieces: category · name · price ·
img · desc · stock · hidden



orders

ref · items · totals · VAT · promo · gift ·
address · method · status



reviews

pid · stars · text · verified · patron



wishlists · carts

per patron, merged on sign-in



resets

one-time codes with expiry & attempt
limit

*Reset the demo: delete db.json — the
catalogue re-seeds and the manager
account is recreated on the next run.*

AUTH

POST signup · login · forgot · reset · logout · GET me · admin/me

CATALOGUE

GET products · POST / PUT /:id / DELETE /:id (manager)

PATRON

GET/PUT cart · GET/PUT wishlist · GET/POST orders · GET reviews, reviews/:pid · POST / DELETE reviews

MANAGER

GET admin/orders · GET admin/users · DELETE admin/users/:email

A Back Office in the Same World



- ◆ Own manager sign-in — verified by the server; patrons are refused
- ◆ Six KPI cards that count up: pieces sold with an improving / declining verdict, revenue, orders & average order, different products sold, stock on hand, new patrons
- ◆ Charts grow into place: units per month with the revenue line, sales-by-collection donut, units by province, users per day
- ◆ Live indicator, greeting and toasts — the same midnight-and-gold as the storefront; keyboard-navigable tabs

PRODUCT MGMT · DONE

REPORTS · DONE

Reports: Data Decides

- ◆ Filter by period (all time · 30 / 90 days · this year · custom months), collection, province and source — every figure recalculates
- ◆ Values: pieces sold + growth verdict, revenue, orders & average order, different products sold (text control), stock on hand & vault value by collection, registered users per day, best sellers
- ◆ Sample history is clearly labelled; live orders from the database merge on top
- ◆ Every chart is drawn by hand on a canvas — no chart library



Add, Edit, Retire, Delete

The screenshot shows the Opaléra product management interface. At the top, there's a header with the Opaléra logo, a greeting 'Good afternoon, Ompha', the date 'SUN, 13 SEPTEMBER', a 'LIVE' status indicator, and buttons for 'VIEW STOREFRONT' and 'SIGN OUT'. The main content area is titled 'EDIT PIECE #9 - HEART NECKLACE'. It contains a form with fields for 'NAME' (Heart Necklace), 'COLLECTION' (Necklace), 'PRICE (R, VAT INCL.)' (1399), and 'STOCK' (3). There's also an 'IMAGE URL' field with a placeholder image and a 'DESCRIPTION' field with a text area. Below the form are 'SAVE CHANGES' and 'CANCEL' buttons. At the bottom, there's a section titled 'ALL PIECES' with a search bar and filters for 'All collections' and 'Any status'. Below this is a table listing three pieces: Sapphire Necklace, Cyan Necklace, and Diamond Necklace, each with its own set of 'EDIT', 'RETIRE', and 'DELETE' buttons.

#	PIECE	COLLECTION	PRICE	STOCK	STATUS	ACTIONS
1	Sapphire Necklace	NECKLACE	R 1 759	16	LIVE	EDIT RETIRE DELETE
2	Cyan Necklace	NECKLACE	R 3 199	16	LIVE	EDIT RETIRE DELETE
3	Diamond Necklace	NECKLACE	R 2 119	9	LIVE	EDIT RETIRE DELETE

- ◆ Add a piece: name, collection, price, stock, photo (with preview) and description → POST /api/products; it gets a product page automatically
- ◆ Edit any piece in place (shown: Heart Necklace) → PUT /api/products/:id; the storefront and its product page update immediately
- ◆ Inline stock control; low-stock and out-of-stock flags; retire hides a piece while keeping its history
- ◆ Delete removes it from the database; search and status filters over all 80 pieces

ADD · 3

EDIT · 5

DELETE · 2

Every Order, Every Patron

OPALÉRA
CONSOLE · BACK OFFICE

Good afternoon, Ompha + SUN, 13 SEPTEMBER LIVE [VIEW STOREFRONT](#) [SIGN OUT](#)

REPORTS INVENTORY & PIECES **ORDERS** PATRONS ENGAGEMENT

Orders

Every order in the database, from every patron, newest first.

Search reference or email Any method 11 of 11 orders

REFERENCES	DATE	CUSTOMER	DESTINATION	ITEMS	VAT	TOTAL	METHOD	STATUS
OPMTZPLSLW	2026/09/13	demo@sample.com	Cape Town, Western Cape	2	R 412	R 3 166	Cash	PAID Invoice
OPMTZPNBN	2026/09/13	demo@sample.com	Cape Town, Western Cape	2	R 328	R 2 518	Cash	PAID Invoice
OPMTYGBRAJ	2026/09/12	demo@sample.com	Cape Town, Western Cape	2	R 328	R 2 518	Cash	PAID Invoice
OPMTLDEIKH	2026/09/09	demo@sample.com	Pretoria, Gauteng	1	R 287	R 2 199	Instant EFT	PAID Invoice
OPMTTTEJZ	2026/09/09	demo@sample.com	Johannesburg, Gauteng	1	R 136	R 1 039	Cash	PAID Invoice
OPMTZPFWD	2026/09/05	demo@sample.com	Johurg, Gauteng	2	R 439	R 3 518	Cash	PAID Invoice
OPMTQSVK3D	2026/09/05	demo@sample.com	Isidore collection	1	R 0	R 1 759	Cash	PAID Invoice
OPMTXGAPF	2026/09/03	demo@sample.com	Johurg, Gauteng	1	R 182	R 1 299	Instant EFT	PAID Invoice
OPMTXGAPF	2026/09/03	demo@sample.com	Johurg, Gauteng	2	R 793	R 6 078	Instant EFT	PAID Invoice
OPMTXGZBHA	2026/09/02	demo@sample.com	Johannesburg, Gauteng	1	R 229	R 1 759	Instant EFT	PAID Invoice
OPMTXGKWSK	2026/09/02	demo@sample.com	Johannesburg, Gauteng	3	R 601	R 4 605	Instant EFT	PAID Invoice

Orders · every order from every patron, newest first: reference, date, customer, destination, items, VAT, total, method, status; search by reference or email, filter by method; open the invoice.

REPORTS · DONE

OPALÉRA
CONSOLE · BACK OFFICE

Good afternoon, Ompha + SUN, 13 SEPTEMBER LIVE [VIEW STOREFRONT](#) [SIGN OUT](#)

REPORTS INVENTORY & PIECES ORDERS **PATRONS** ENGAGEMENT

Patrons

Registered accounts and when they joined; the "Registered users per day" chart on the Reports tab follows the same period filter.

ACCOUNTS 2 registered · 1 manager

NAME	EMAIL	ROLE	JOINED	ORDERS
Ompha Manager	manager@opalera.co.za	MANAGER	09 Sept 2026	0
Demo Patron	demo@sample.com	PATRON	02 Sept 2026	11

SIGN-UPS BY DAY

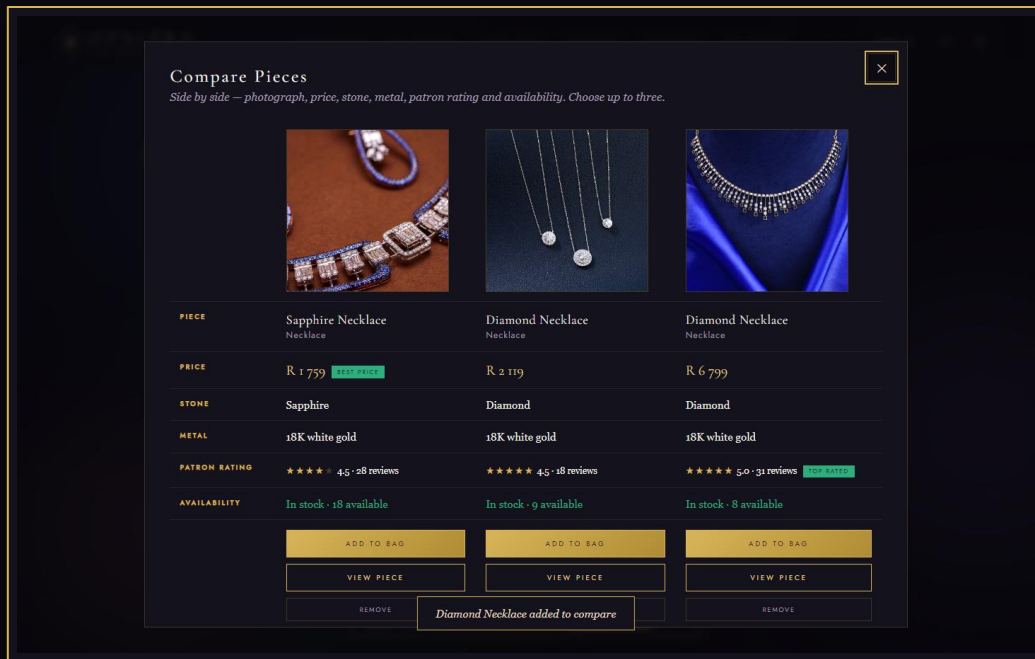
DAY	NEW ACCOUNTS
Wed, 09 Sept 2026	1
Wed, 02 Sept 2026	1

Patrons · every account with role, sign-up date and order count; sign-ups by day beside it. Managers can remove a patron account (managers are protected).

USER MGMT · DONE

Compare, Track, Scan

- ◆ Compare pieces — tick up to three; a side-by-side table of photo, price (“Best price”), stone, metal, rating (“Top rated”) and availability, with add-to-bag in place
- ◆ Order tracking — a live timeline per order, with status chips and ETA on the account page
- ◆ QR code on every invoice — printed invoices open the order's tracking page from a phone
- ◆ Also: the Studio customiser, and a motion system that respects reduced-motion



Built to Be Trusted



64 automated checks — local and live

Site & static blocking, sign-up / login / reset, cart, wishlist, orders with server-side totals, the stock rule, reviews, roles, product CRUD and validation, then cleanup of its own test data — all passing against localhost and against opalera.onrender.com.



Browser-verified pages

Every page loaded with zero console errors; compare, tracking, invoice QR, account and the console exercised end to end; keyboard navigation checked.



Security by default

script password hashes with per-user salt, constant-time comparison, httpOnly SameSite cookies, server-recomputed totals and stock rules, the backend folder never served over HTTP, role checks on the server, a curated seed database with no tokens.



Accessible & considerate

Reduced-motion switch, WAI-ARIA tabs with arrow keys, live regions, focus rings, colour contrast checked; a multi-agent review found and fixed 29 defects before release.

Live on Render, Runs Anywhere Node Does

RUN IT LOCALLY

```
node opalera-backend/server.js
```

- ◆ Then open <http://localhost:3000> — it redirects to the storefront
- ◆ One server hosts the whole site and its API; no installs, no build step
- ◆ Data lives in `opalera-backend/data/db.json` (or `DATA_DIR`), created on first run

DEMO ACCOUNTS

Patron demo@example.com / `demo123`

Manager

manager@opalera.co.za / `Opalera2026`



Live site

opalera.onrender.com

Storefront /pages/jewellery_Html.html · Console </pages/admin.html>



Source & workflow

github.com/Thrivi17/Opalera

Both members commit to main as collaborators; every push redeploys the live site within minutes. The seed database, docs and presentation live in the repo.

ALL PIECES 80

NECKLACES 30

EARRINGS 30

PENDANTS 30

WISHLIST 0

THE STUDIO +

Search the archive

Every Price

Curated Order

SHOWING 80 OF 80 PIECES

Merci.

OPALÉRA — *every promise from the proposal, delivered on the page.*

QUANTUM CODERS · IFM2B10 · FINAL PRESENTATION · SEPTEMBER 2026

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